

Question	Answer	Marks
8(a)	Every point mass attracts every other point mass with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.	A2
(b)(i)	$a_c = r \omega^2$ $a_c = (6.37 \times 10^6) \left(\frac{2\pi}{24 \times 60^2} \right)^2$ $a_c = 0.0337 \text{ m s}^{-2}$	C1 A1
(b)(ii)	$g = \frac{GM}{r^2} = \frac{6.67 \times 10^{-11} (6.0 \times 10^{24})}{(6.37 \times 10^6)^2} = 9.863 \text{ m s}^{-2}$ $a_{\text{freefall}} = g - a_c$ $a_{\text{freefall}} = 9.863 - 0.0337$ $a_{\text{freefall}} = 9.83 \text{ m s}^{-2}$ A small fraction of the gravitational force acted by the earth on any body near the equator is to provide for the centripetal acceleration; at the pole, the centripetal acceleration is zero and the acceleration of free-fall is entirely due to the pull of earth.	B1 A1 A1 A1
(c)(i)	Gravitational force provides for centripetal force: $\frac{GM_E m}{r^2} = mr \left(\frac{2\pi}{T} \right)^2$ $T^2 = \frac{4\pi^2 (6.37 \times 10^6 + 408 \times 10^3)^3}{(6.67 \times 10^{-11}) (5.97 \times 10^{24})} \quad \text{(Note: use of correct radius)}$ $T = 5556 \text{ s}$ $T = \frac{5556}{3600} = 1.5 \text{ hr (shown)}$ First mark for formulation of relationship Second mark for correct substitution of values and the calculation of the period in seconds.	M1 M1
(c)(ii)	The Kepler's Law holds for scenarios where the gravitational force provides entirely for the centripetal force hence it holds for the satellite in orbit. However, for student on Earth's surface, there are other forces acting on him (normal contact, etc) and the resultant (centripetal) force is not the gravitational force	A1 A1
(c)(iii)	Both astronaut and space station are undergoing circular motion and have the <u>same centripetal acceleration</u> provided by their weights. There is <u>no contact force acting on him by his surrounding</u> and hence he perceived himself to be weightless.	B1 B1

	Alternative: if a clear account of what true weightlessness mean and how the fact that gravity must have provided the centripetal acceleration, one mark will be awarded.	
(d)(i)	1. $\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{(7 \times 10^{22})(1 \times 10^3)} = 9.5 \times 10^{-60} \text{ m}$ 2. $\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{(9 \times 10^{-31})(2 \times 10^7)} = 3.7 \times 10^{-11} \text{ m}$ Both correct – 3 marks One correct – 2 marks Method and substitution correct but answers wrong – 1 mark	B1 A1 A1
(d)(ii)	For the Moon, its de Broglie's wavelength is very much smaller than the orbital radius. For the electron, its de Broglie's wavelength is comparable to its orbital radius. Thus the wave-like properties for electron cannot be neglected for its motion around the nucleus while that of Moon can be neglected.	B1 B1 B1
	Max Marks	20